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EFFECT OF PROCESS PARAMETERS ON AEROSOL JET PRINTING OF MULTI-LAYER CIRCUITRY

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ABSTRACT

Printing technologies such as Aerosol Jet provides the freedom of miniaturizing interconnects and producing fine pitch components. Aerosol Jet, a direct printing technique replaces the traditional steps of manufacturing a printed circuit board such as lithography or etching, which are quite expensive, and further allowing the circuits to be fabricated onto all kinds of substrates. Wide impact areas range from healthcare to wearables to future automotive applications. The aerosol jet printer from Optomec utilized in this study, consists of two types of atomizers depending on ink viscosity. One is Ultrasonic Atomizer which supports ink with viscosity range of 1-5cP, and another is Pneumatic Atomizer with large range of suitable viscosity 1-1000cP. This paper focuses on utilizing the aerosol jet printing using both the atomizers to develop process parameters to be able to successfully print bi-material, multi-layer circuitry. The insulating material between two conductive lines used in the paper is of very high viscosity of 350cP, suitable for Pneumatic atomizer and Silver Nano-particle ink with the viscosity suitable for Ultrasonic atomizer as a conductive ink. A statistical modeling approach is presented to predict the attributes such as micro-via diameter before starting the print process, enabling us to pre-adjust the dimensions in CAD for the desired output. Process parameters to obtain a fine print with good electrical properties and better dimensional accuracy are developed. Importance of pre-cleaning the substrate is discussed, in addition to the printing process efficiency gauged as a function of process capability index and process capability ratio.

Keywords: Aerosol Jet, multi-layer, micro via, dielectric, process capability ratio and index, regression model

NOMENCLATURE

AJP	Aerosol Jet Printing
UA	Ultrasonic Atomization
PA	Pneumatic Atomization
cP	Centipoise, Unit of Viscosity
C_p	Process Capability Ratio
C_{pk}	Process Capability Index
sccm	Flow Rate in Standard Cubic Centimeters per Minute

1. INTRODUCTION

Digital printing technologies are rapidly gaining importance for manufacturing electronic devices with their extremely low fabrication cost and be able to print on flexible and conformal substrates such as polyimide rather than conventional FR-4. One of the technologies, Aerosol Jet, works wonders when it comes to the wide selection of substrates and materials to be able to print. AJP process utilizes aerodynamic focusing to focus a collimated mist of Nano-particles onto a substrate using an inert carrier gas known as sheath gas, such as nitrogen. To account for the wide array of materials, the process further breaks down to two different types of atomization: UA for low viscosity materials (1 to 5cP) which utilizes ultrasonic waves to atomize; and PA for low and high viscosity materials (1 – 1000cP) which utilizes an inert carrier gas to atomize, thereby suspending the Nano-particles into air which travels through a mist tube and gets deposited. Both of the processes are depicted in FIGURE 1-FIGURE **2**.

Aerosol-jet printing process requires the control of multiple parameters simultaneously for a fine print such as atomizer flow rate, sheath flow rate, nozzle size, print speed, base temperature, UA current, print height (distance between nozzle exit and the

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substrate), number of passes. Furthermore, for desired pattern, the print process requires a toolpath file to follow which can be generated from a 2D CAD model, implying there is no specification of the line width. The desired line width is obtained by changing the above-mentioned parameters accordingly, which can be time consuming since each material is of a different viscosity and varying metal and solvent composition.

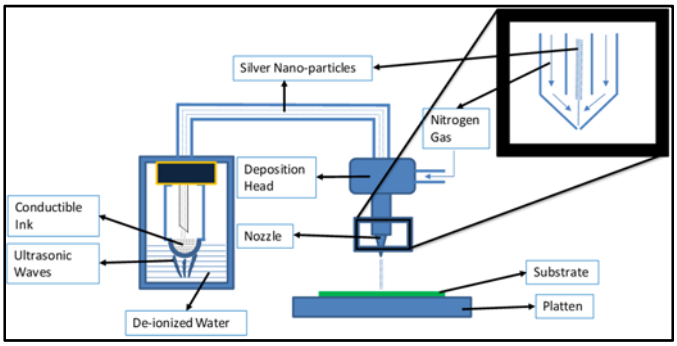

Figure 1: Principle of Ultrasonic Atomization [1]

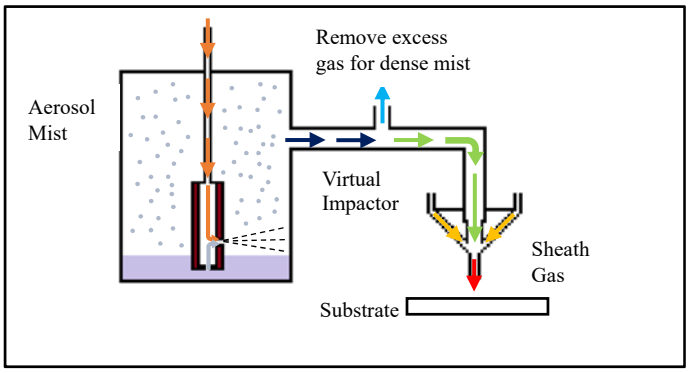

Figure 2: Principle of Pneumatic Atomization [2]

Much work has been done in the field of AJP technology. A basic overview and how to utilize Aerosol Jet to print functional circuit board with different types of components and attachment is discussed by researchers [3]. Aerosol jet printing has been used for fabrication of silicon solar cells [4]. In addition, the aerosol jet printing has been used for printing on conformal surfaces including multi-material printing, non-planar and orthogonal surface printing [5]. Factors of importance in multilayer circuits have been discussed by researchers [6]. AJP technology has also been used to demonstrate coplanar waveguides up to 50 GHz on polyimide substrate [7]. Micro-inductor fabrication has also been achieved with multi-material consisting of two Ag metal lines separated by polyimide dielectric using aerosol jet [8].

In this paper, a statistical approach is presented for prediction of physical attributes of multilayer circuits micro-via diameter for the dielectric and line widths and resistance for the metal conductive line before starting the print process. The model can also be used to pre-adjust the CAD file dimensions to obtain the desired characteristics. Next, process parameters are developed for a multi-layer print consisting of metal Nano-particle ink as a conductive path and an electronic adhesive as a dielectric to provide the insulation between two conductive paths. Finally, printing process efficiency is gauged as a function of process capability index (C_{pk}) and process capability ratio (C_p).

2. TEST DESIGN

The test design is modeled to create a multi-layer print which utilizes both the atomizers, UA and PA for the metal conductive line and dielectric respectively. The modeling is done in AutoCAD, and shown in Figure 3. The composition is as follows:

- Layer 1 (Ag metal line) is shown in red color, which consists of two 16 mm lines from the center of one pad to another, four square landing pads for dielectric of 0.4 mm, and four square pads at the end of 3 mm.
- Layer 2 (dielectric) is shown in green color, which consists of four vias with diameter of 0.3 mm (~12mils), and four filled insulating square pads of 1.2 mm localized to the region of crossover and separating the metal lines.
- Layer 3 (Ag metal line) is shown in blue color, which consists of four circular pads of 0.6 mm to fill up the via, four rectangular landing pads of 0.5 mm x 0.75 mm for the metal line on top of dielectric, and similar to layer 1, two 16 mm lines from the center of one pad to another and four square landing pads for dielectric of 0.4 mm.

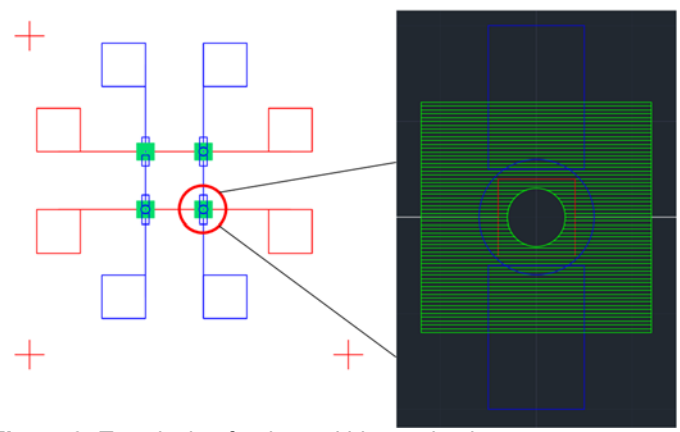

Figure 3: Test design for the multi-layer circuitry

2.1 For Statistical Modeling

Different test design is modeled to create the statistical model for use in prediction of the physical attributes. In the case of dielectric, prediction is done for via diameter, whereas for metal conductive line, it is for the line width and resistance. Figure 4 shows the design for the dielectric, which consists of multiple columns of via diameters representing different combination of tests and ten rows of the diameters at the same tests to check for repeatability. Similarly, Figure 5 shows the design for metal conductive lines consisting of multiple columns representing different combination of tests and five rows at the same test for repeatability.

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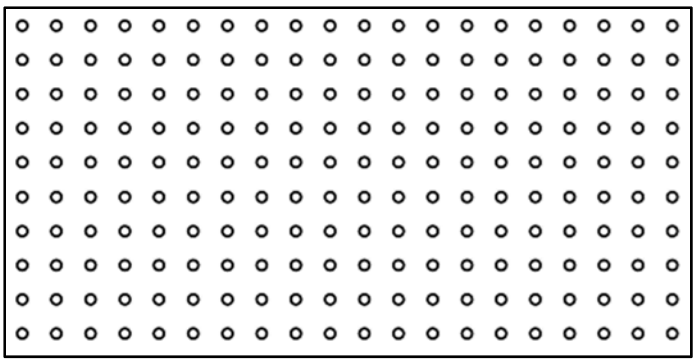

Figure 4: Test design for the statistical model for dielectric via diameter prediction

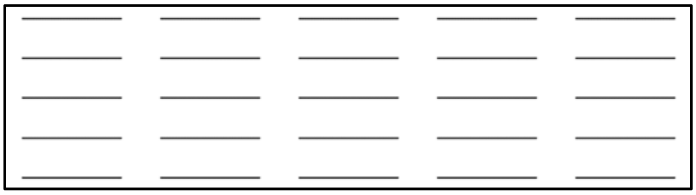

Figure 5: Test design for the statistical model for conductive line width and resistance prediction

3. MATERIALS AND METHODS

The materials selected in this paper are chosen keeping in mind the two atomization processes namely ultrasonic and pneumatic which supports low and high viscosity materials respectively. Therefore, for conductive paths, low viscosity silver metal Nano-particle ink, references as Ink A, is used using UA, and for insulation, high viscosity electronic adhesive, referenced as Ink 2, as a dielectric is used using PA. The properties and composition for the two materials are listed in Table 1 and Table 2. The following section discusses the procedure for the statistical modeling and printing process efficiency as a function of process capability index (C_{pk}) and process capability ratio (C_p).

Table 1: Typical Properties of Ink 2 Dielectric

Typical Properties of Ink 2	
Viscosity @ 25°C (cP)	300
Modulus of Elasticity (psi)	160,000
Hardness – Shore D	85
Dielectric Constant (1 MHz)	4.04
Dielectric Strength (Volts)	980

Table 2: Typical Formulation of Ag Nanoparticle Ink

Typical Properties of Ink 1	
Viscosity (can be adjusted) (cP)	1 – 30
Particle Size (nm)	~8 – 15
Silver Concentration (wt. %)	~20 – 60
Curing Temperature (°C)	120 – 300

3.1 For Statistical Modeling

As discussed above, AJP process requires control of several parameters simultaneously to obtain a fine print. Considering the number of parameters, it would be unwise to build a statistical model which is unreasonably time consuming. Therefore, a screening process is done using our knowledge of AJP process to separate out the effects playing an important role in the print quality. The levels of parameters that are important but have little impact on the print quality have been reduced in comparison with the main important effects. Additionally, for a baseline model, only the main effects and two-way interactions are considered. Note that this modeling approach is valid only for the materials discussed, and not a generalized model since every material is different, in terms of its viscosity, composition, material properties and wetting characteristics.

3.1.1 For Ink 2 using PA

To perform the study, a list of controlled and uncontrolled factors is made in addition to the output response. Table 3 provides the list of controlled factors with their levels considered, while Table 4 provides the uncontrolled or fixed factors. Table 5 lists the output response.

Table 3: List of Controlled Factors

Controlled Factors			
PA_MFC (sccm)	Sheath (sccm)	Δ (PA_MFC – Exhaust) (sccm)	Print Speed (mm/s)
3 Levels $\begin{Bmatrix}\mathbf{1150}\\ \mathbf{1200}\\ \mathbf{1250}\end{Bmatrix}$	3 Levels $\begin{Bmatrix}50\\ 75\\ 100\end{Bmatrix}$	3 Levels $\begin{Bmatrix}40\\ 50\\ 60\end{Bmatrix}$	3 Levels $\begin{Bmatrix}0.3\\ 0.5\\ 0.6\end{Bmatrix}$

Table 4: List of Uncontrolled Factors

Uncontrolled Factors	
Platen Temperature (°C)	Room temperature
Atomization	Pneumatic
Dielectric Material	Adhesive
Substrate	Glass
Number of Passes	2
Nozzle (μm)	200
Substrate Pre-cleaning	5min IPA submerged and Ultrasonic cleaned with DI water
Curing	UV Cure on the Fly

Table 5: Output Response

Output Response	Via Diameter

Using the information from above, DOE (Design-of-Experiment) matrix is created which provides us the combinations to perform the experiments on. With that matrix, and design discussed previously, the study is carried out. Ten rows of via diameters are printed at the same test condition, after which parameters are changed and so on. Note that this is valid for only Ink 2 with PA. In addition, the range of the controlled factors are chosen to be very close to each other to get a

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connected line for the diameter measurement. Increasing the range gives us unconnected line at some levels, which can produce errors in the model.

3.1.2 For Ink 1 using UA

Similar to section 3.1.1, study was done to predict the line width and the resistance. Main effects and only two-way interactions are considered. Since this is a low viscosity metal ink, it is easier to have more controllable factors as compared to in the case of dielectric. Table 6-Table 8 lists the controlled, uncontrolled factors, and the output responses.

Table 6: List of Controlled Factors

Controlled Factors				
UA_MFC (sccm)	Sheath (sccm)	Print Speed (mm/s)	Standoff Height (mm)	Number of Passes
4 Levels $\begin{Bmatrix}\mathbf{10}\\\mathbf{15}\\\mathbf{20}\\\mathbf{25}\end{Bmatrix}$	3 Levels $\begin{Bmatrix}50\\100\\150\end{Bmatrix}$	3 Levels $\begin{Bmatrix}3\\5\\7\end{Bmatrix}$	3 Levels $\begin{Bmatrix}2\\3\\4\end{Bmatrix}$	3 Levels $\begin{Bmatrix}3\\5\\10\end{Bmatrix}$

Table 7: List of Uncontrolled Factors

Uncontrolled Factors	
Platen Temperature (°C)	55
UA Current	
Chiller Temperature (°C)	14
Atomization	Ultrasonic
Conductive Ink Material	Ag Nano-particle
Substrate	Glass
Substrate Pre-cleaning	5min IPA submerged and Ultrasonic cleaned with DI water
Sintering	*Layer 1:* 100°C for 30 mins *Layer 3:* 300°C for 30 mins

Table 8: Output Responses

Output Responses	▪ Line Width

Using the information above, similar to section 3.1.1, DOE matrix is created for the combination of test parameters. With the design provided in section 2.1, the study was carried out. The results are valid for Ink 1 with UA – however, the trends exhibited in various variables are more general to other materials as well.

3.2 Printing Process Efficiency

This section aims at using a statistical approach to understand if a process is robust. Response variables from the modeling study can be used as potential variables to use in the calculation of process capability index (C_{pk}) and process capability ratio (C_p). These process capability variables will help to determine if the process is under control for a reliable use. C_p – C_{pk} formulations will be established for via diameter. These formulations for the process parameters of AJP for the ink used have already been established [1].

4. RESULTS AND DISCUSSION

Process study is developed for the two materials discussed above and a statistical approach is presented to predict the physical attributes. Using the results from the statistical modeling, response factors are gauged as a function of process capability variables.

4.1 Process Study

Process study consists of creating a micro-via with "optimum" set of AJP parameters, curing conditions, and most importantly, pre-cleaning of the substrate. Figure 6 below shows a complete micro-via with the AJP parameters listed in Table 9Table 10.

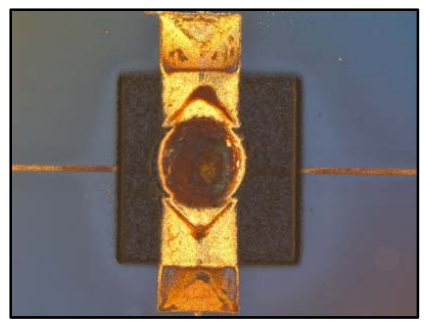

Figure 6: Micro-Via

Table 9: Set of Process Parameters for the Micro-Via

Process Parameters for Dielectric	
Carrier Flow Rate (sccm)	1200
Sheath (sccm)	50
Exhaust or Δ (sccm)	1150 or 50
Print Speed (mm/s)	0.5
Standoff Height (Distance b/w Nozzle tip and Substrate) (mm)	2
Nozzle (μm)	200
Number of Passes	2
Curing	UV Cure on the Fly
Pre-Cleaning	5min IPA submerged and Ultrasonic cleaned with DI water

Table 10: Set of Process Parameters for the Metal Ink

Process Parameters for Ag Nanoparticle Ink	
Carrier Flow Rate (sccm)	18
Ultrasonic Current (Amps)	0.54 – 0.56
Sheath (sccm)	100
Print Speed (mm/s)	5
Standoff Height (Distance b/w Nozzle tip and Substrate) (mm)	2
Nozzle (μm)	200
Number of Passes	5
Sintering	*Layer 1:* 100°C for 30 mins *Layer 3:* 300°C for 30 mins

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Process Parameters for Ag Nanoparticle Ink	
Pre-Cleaning	5min IPA submerged and Ultrasonic cleaned with DI water

4.2 Effect of UV Intensity and Sintering on Micro-Via

This section discusses the effect of different UV intensities and uneven sintering conditions for Layer 1 and Layer 3 used to cure Ink 2 and the Ag metal conductive line respectively. The UV system used is from Panasonic, model Aicure UJ35 [9]. It consists of four individually controlled UV-LED heads with cylindrical lenses to spot cure. Fixtures are designed to hold the LED in place on the AJP machine to UV cure while the toolpath is running.

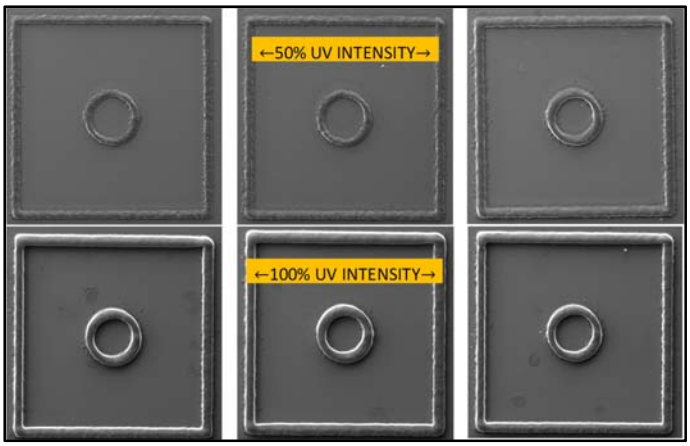


Figure 7: Effect of UV Intensity on the Via

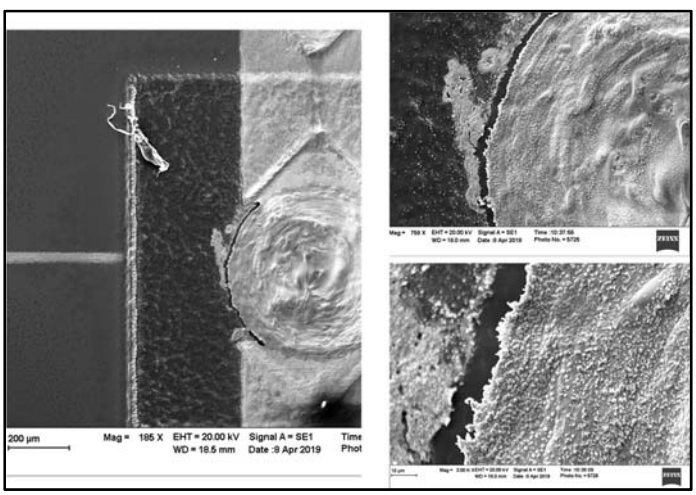


Figure 8: Effect of over-sintering the Via (200°C/60mins for Layer 1; 300°C/60mins)

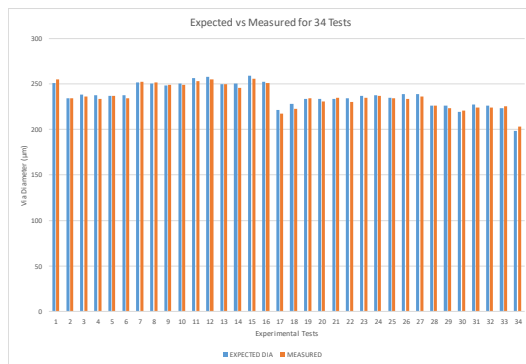
From Figure 7, a clear distinction can be seen when Ink 2 is cured at two different intensities i.e. 50% and 100%. Similarly, curing the metal line with proper set of temperatures is just as important since Layer 1, which is sintered at specific temperature will be exposed to another sintering condition when Layer 3 is getting cured. Figure 8 shows an interface crack between Ag and dielectric when the curing conditions were 200°C/60 mins for Layer 1 and 300°C/60 mins. The best combination of sintering parameters were determined by varying the temperature and time, and measuring their effect on the line resistance and shear load to failure. Using the results, Layer 1 was sintered at 100°C/30mins and Layer 2 at 300°C/30mins.

4.3 Statistical Approach

A statistical significant regression model for any type of material gives us the ability to predict the response variable before running an experiment. Here we develop a model considering screened-out controlled factors and a response variable. For both, dielectric and Ag metal ink, controlled factors and output response are listed in section 3.1. Later on, statistical measures such as process capability variables (C_p – C_{pk}) will be used to determine if the AJP process is reliable for consistency.

4.3.1 Model for Ink 2 Dielectric

From the modeled DOE and using a D-optimality measure, 34 combination of tests were performed using the dielectric ink and the via diameter was measured for all of them. Figure 9 below shows a comparison graph for the via diameter expected vs. measured for those tests. The expected diameter is the one modeled through AutoCAD. The measured diameters were recorded as the response to input into the statistical model.



- Expected via diameter:
  - AutoCAD defined dia (300µm) – Line Width
- Measured Via diameter

Figure 9: Via Diameter measurements for the 34 Tests performed

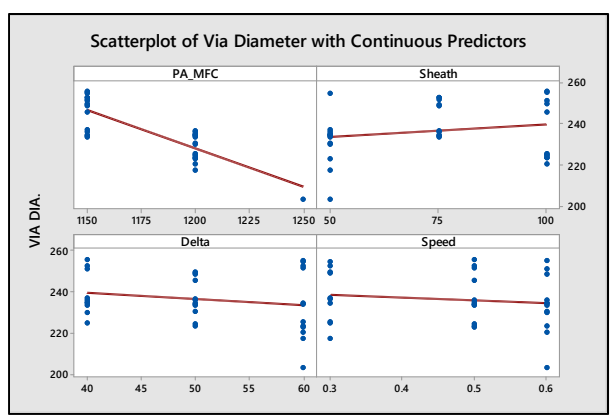


Figure 10: Fitted Plots for Main Effects (Dielectric)

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With the list of controlled factors and their levels discussed in section 3.1.1, in addition to the measured response from Figure 9, a linear regression model was developed. Fitted plots for the main effects are shown in Figure 10. Only main effects and their two-way interactions are considered for the baseline model. With the 90% confidence interval, from Figure 11, we can see the significant factors which have *p*-values <0.1. Additionally, the fact that an important main variable "speed" not being significant can be due to the levels considered in the model. The reason for those levels comes from the fact that how the material does not form a consistent line at higher print speeds and depositing too much at lower speed. Upon further evaluation, we see there is high multicollinearity issues as shown in red in Figure 12, which can result in misinterpretation of the variables. Steps such as standardizing the variables, transforming the dataset are taken to account for the multicollinearity issues discussed later.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	10	4548.64	454.864	13.20	0.000
PA_MFC	1	134.96	134.958	3.92	0.060
Sheath	1	369.45	369.452	10.72	0.003
Delta	1	294.61	294.613	8.55	0.008
Speed	1	18.06	18.056	0.52	0.477
PA_MFC*Sheath	1	376.06	376.058	10.91	0.003
PA_MFC*Delta	1	291.19	291.195	8.45	0.008
PA_MFC*Speed	1	18.74	18.739	0.54	0.468
Sheath*Delta	1	6.14	6.138	0.18	0.677
Sheath*Speed	1	53.34	53.339	1.55	0.226
Delta*Speed	1	27.72	27.721	0.80	0.379
Error	23	792.85	34.472		
Total	33	5341.49			

Figure 11: ANOVA for Baseline Model (Dielectric)

To make sure the model is not violating any assumptions for the regression model, residual plots are plotted in Figure 13. The residuals appears to be normally distributed with constant variance with an outlier present. From the histogram, data does appear to be little skewed towards right, and from the residuals versus order graph, there is no correlation in the residuals for the 34 tests performed.

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-555	399	-1.39	0.177	
PA_MFC	0.678	0.343	1.98	0.060	88.47
Sheath	7.07	2.16	3.27	0.003	2021.04
Delta	16.19	5.54	2.92	0.008	1947.36
Speed	-269	372	-0.72	0.477	2014.56
PA_MFC*Sheath	-0.00623	0.00189	-3.30	0.003	2134.05
PA_MFC*Delta	-0.01365	0.00470	-2.91	0.008	2065.68
PA_MFC*Speed	0.234	0.318	0.74	0.468	2095.33
Sheath*Delta	0.00265	0.00629	0.42	0.677	58.86
Sheath*Speed	0.501	0.403	1.24	0.226	30.11
Delta*Speed	-1.00	1.11	-0.90	0.379	60.66

Figure 12: Multicollinearity Issues (Dielectric)

To account for the multicollinearity problems, variables are standardized i.e. subtract from the mean and dividing by the standard deviation. Additionally, removal of outlier was also done. Modified residual plots are shown in Figure 14.

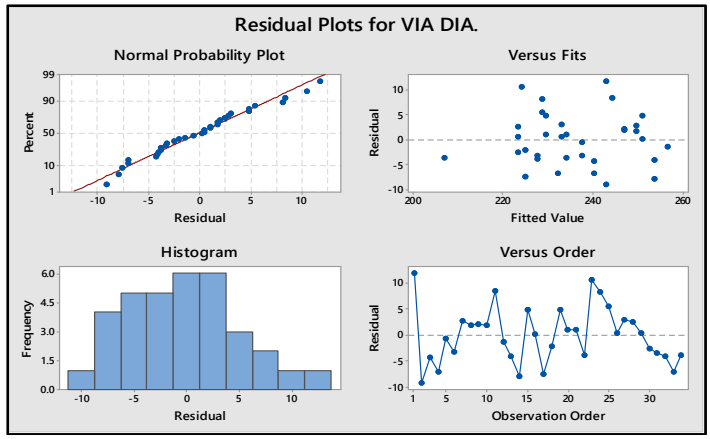

Figure 13: Residual Plots for Model Assumptions (Dielectric)

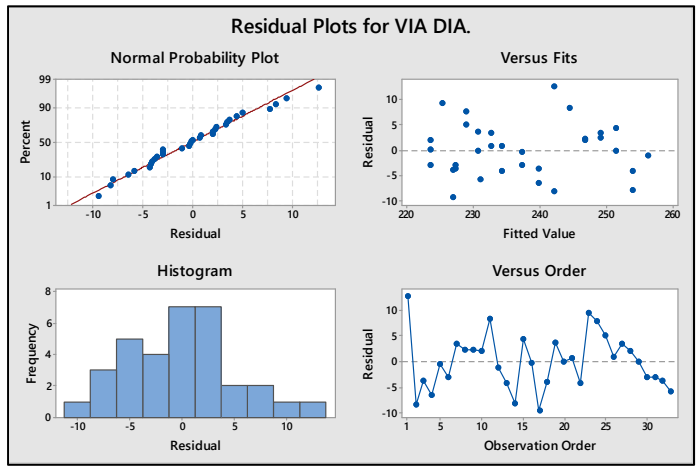

Figure 14: Residual Plots after Model Transformation (Dielectric)

Stepwise Selection of Terms

α to enter = 0.15, α to remove = 0.15
The stepwise procedure added terms during the procedure in order to maintain a hierarchical model at each step.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	5	3331.38	666.28	20.37	0.000
PA_MFC	1	2595.11	2595.11	79.34	0.000
Sheath	1	147.30	147.30	4.50	0.043
Delta	1	11.34	11.34	0.35	0.561
PA_MFC*Sheath	1	424.87	424.87	12.99	0.001
PA_MFC*Delta	1	187.74	187.74	5.74	0.024
Error	27	883.14	32.71		
Total	32	4214.52			

Continuous predictor standardization
Subtract the mean, then divide by the standard deviation

Predictor	Mean	StDev
PA_MFC	1175.76	25.3760
Sheath	74.24	21.1808
Delta	50.30	8.0951
Speed	0.47	0.1232

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
5.71917	79.05%	75.16%	67.89%

Figure 15: Stepwise Variable Selection with R^2 (Dielectric)

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Residual plots for the transformed dataset being satisfied, a stepwise regression analysis was performed to remove the insignificant variables from the model. Level of significance for keeping the variable and removing it was kept to be at 0.15 for making the variable easier to enter the model while not being very harsh to remove it. Figure 15 shows the significant factors left in the model with their *p*-values. With these significant variables, the total variation explained by the model is about 79%, with adjusted R^2 of 75%. The regression equation to predict the via diameter using the dielectric with pneumatic atomization of the aerosol jet printer is shown in Figure 16 i.e.

$$\boxed{\begin{aligned}(\text{Via Dia.}) = -672 + 0.77(\text{PA_MFC}) + 8.29(\text{Sheath}) \\ +14.16(\text{Delta}) - 0.0069(\text{PA_MFC} * \text{Sheath}) \\ -0.012(\text{PA_MFC} * \text{Delta})\end{aligned}}$$

Where,

PA_MFC:	Pneumatic Carrier Gas Mass Flow Rate
Sheath:	Nitrogen gas to focus the aerosol onto substrate
Delta:	Net Mass flow rate (Carrier Gas Rate – Exhaust)

Coded Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	237.613	0.997	238.34	0.000	
PA_MFC	-9.02	1.01	-8.91	0.000	1.00
Sheath	2.17	1.02	2.12	0.043	1.02
Delta	-0.60	1.02	-0.59	0.561	1.02
PA_MFC*Sheath	-3.74	1.04	-3.60	0.001	1.02
PA_MFC*Delta	-2.49	1.04	-2.40	0.024	1.02

Regression Equation in Uncoded Units

```
VIA DIA. = -672 + 0.770 PA_MFC + 8.29 Sheath + 14.16 Delta - 0.00696 PA_MFC*Sheath
           - 0.01211 PA_MFC*Delta
```

Figure 16: Regression Model for the Dielectric

4.3.2 Model for Ink 1

Similar to section 4.3.1, a regression modeling approach was adopted to predict the line width for the metal conductive ink with UA used in this article. From the modeled DOE and using a optimality measure, 45 combination of tests were performed and the line width of each of those tests was measured. Figure 17 shows some of the lines printed out of the 45 tests combination.

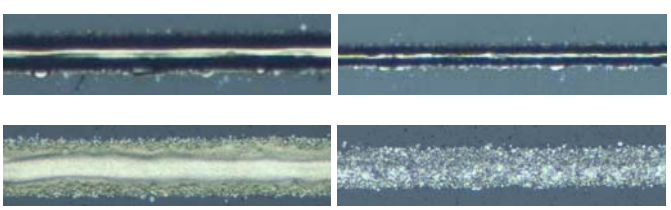

Figure 17: Examples of Some Metal Conductive Printed Lines

With the list of controlled factors and their levels discussed in section 3.1.2, with line width as a response variable, a linear regression model was developed. Fitted plots for the main effects are shown in Figure 18. For the base design, main effects with their two-way interactions are considered for ANOVA. With the 90% confidence interval, significant variables with the *p*-values < 0.1 can be seen in the Figure 19. Since the whole aerosol printing process depends on all the parameters working together, there is high multicollinearity, which can be seen in Figure 20.

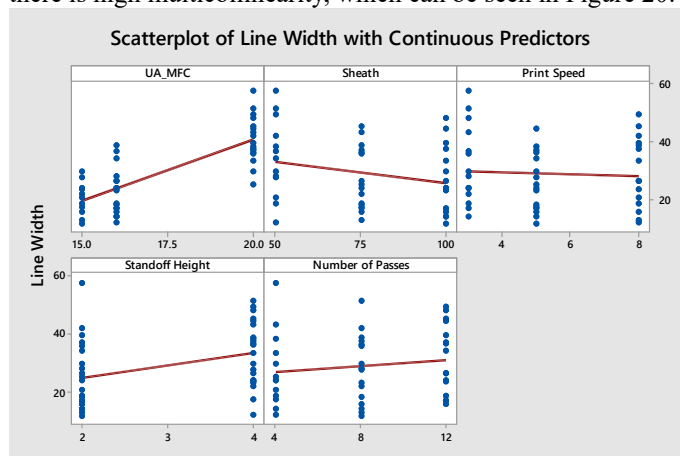

Figure 18: Fitted Plots for the Main Effects (Conductive Ink)

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	15	5355.73	357.048	12.35	0.000
UA_MFC	1	33.48	33.476	1.16	0.291
Sheath	1	47.50	47.503	1.64	0.210
Print Speed	1	12.16	12.161	0.42	0.522
Standoff Height	1	35.49	35.489	1.23	0.277
Number of Passes	1	166.95	166.952	5.78	0.023
UA_MFC*Sheath	1	0.31	0.306	0.01	0.919
UA_MFC*Print Speed	1	17.01	17.012	0.59	0.449
UA_MFC*Standoff Height	1	5.17	5.173	0.18	0.675
UA_MFC*Number of Passes	1	70.01	70.010	2.42	0.130
Sheath*Print Speed	1	62.70	62.697	2.17	0.152
Sheath*Standoff Height	1	111.52	111.522	3.86	0.059
Sheath*Number of Passes	1	10.29	10.293	0.36	0.555
Print Speed*Standoff Height	1	10.80	10.800	0.37	0.546
Print Speed*Number of Passes	1	188.73	188.730	6.53	0.016
Standoff Height*Number of Passes	1	64.07	64.072	2.22	0.147
Error	29	838.11	28.900		
Total	44	6193.84			

Figure 19: ANOVA for Baseline Model (Conductive Ink)

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	42.5	51.2	0.83	0.413	
UA_MFC	2.56	2.38	1.08	0.291	41.23
Sheath	-0.514	0.401	-1.28	0.210	100.12
Print Speed	-2.67	4.12	-0.65	0.522	108.23
Standoff Height	-8.99	8.11	-1.11	0.277	102.43
Number of Passes	-6.33	2.64	-2.40	0.023	110.74
UA_MFC*Sheath	-0.0025	0.0247	-0.10	0.919	142.39
UA_MFC*Print Speed	-0.177	0.231	-0.77	0.449	117.85
UA_MFC*Standoff Height	0.168	0.398	0.42	0.675	93.69
UA_MFC*Number of Passes	0.241	0.155	1.56	0.130	128.29
Sheath*Print Speed	0.0349	0.0237	1.47	0.152	35.36
Sheath*Standoff Height	0.0852	0.0433	1.96	0.059	28.46
Sheath*Number of Passes	-0.0089	0.0150	-0.60	0.555	33.77
Print Speed*Standoff Height	-0.262	0.428	-0.61	0.546	20.90
Print Speed*Number of Passes	0.396	0.155	2.56	0.016	24.92
Standoff Height*Number of Passes	0.415	0.279	1.49	0.147	20.48

Figure 20: Multicollinearity Issues (Conductive Ink)

Residuals are also plotted to make sure if the regression model is not violating any assumptions of regression. From Figure 21, residuals seems to be having a "S" shaped normal distribution with constant variance. The histogram appears to be little right skewed, and no correlation in the order of experiments performed.

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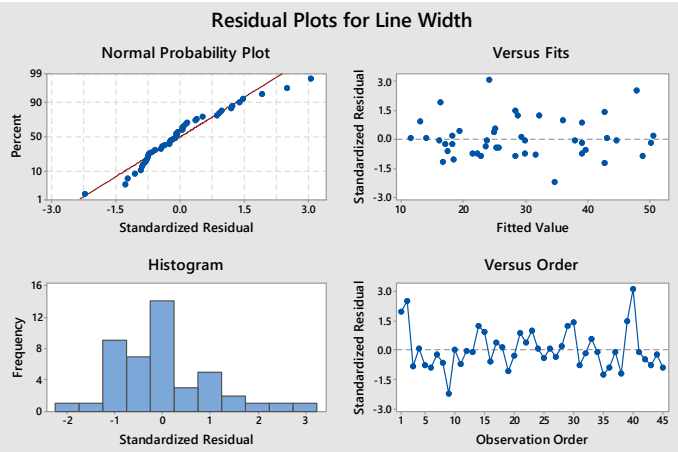

Figure 21: Residual Plots for Model Assumptions (Conductive Ink)

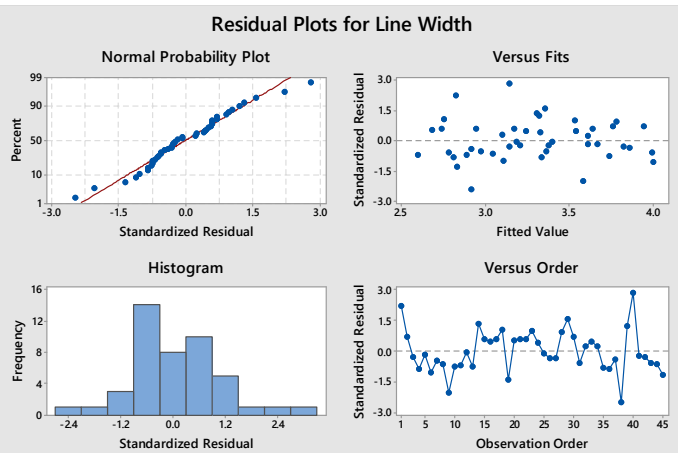

Figure 22: Residuals after Model Transformation (Conductive Ink)

Analysis of Variance for Transformed Response

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	9	6.61763	0.73529	19.62	0.000
UA_MFC	1	3.47330	3.47330	92.67	0.000
Sheath	1	0.71871	0.71871	19.17	0.000
Print Speed	1	0.07677	0.07677	2.05	0.161
Standoff Height	1	0.41431	0.41431	11.05	0.002
Number of Passes	1	0.15192	0.15192	4.05	0.052
UA_MFC*Sheath	1	0.09969	0.09969	2.66	0.112
Sheath*Standoff Height	1	0.18269	0.18269	4.87	0.034
Print Speed*Number of Passes	1	0.52542	0.52542	14.02	0.001
Standoff Height*Number of Passes	1	0.12401	0.12401	3.31	0.077
Error	35	1.31185	0.03748		
Total	44	7.92948			

Continuous predictor standardization
Subtract the mean, then divide by the standard deviation

Predictor	Mean	StDev
UA_MFC	17.2667	2.1888
Sheath	76.6667	20.2260
Print Speed	5.3778	2.0482
Standoff Height	2.9778	1.0111
Number of Passes	8.2667	3.2362

Model Summary for Transformed Response

S	R-sq	R-sq(adj)	R-sq(pred)
0.193601	83.46%	79.20%	72.47%

Figure 23: Stepwise Variable Selection with R^2 (Conductive Ink)

Coded Coefficients for Transformed Response

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	3.2777	0.0291	112.78	0.000	
UA_MFC	0.2872	0.0298	9.63	0.000	1.04
Sheath	-0.1321	0.0302	-4.38	0.000	1.07
Print Speed	-0.0426	0.0298	-1.43	0.161	1.04
Standoff Height	0.0997	0.0300	3.32	0.002	1.06
Number of Passes	0.0600	0.0298	2.01	0.052	1.04
UA_MFC*Sheath	0.0558	0.0342	1.63	0.112	1.35
Sheath*Standoff Height	0.0674	0.0305	2.21	0.034	1.07
Print Speed*Number of Passes	0.1260	0.0337	3.74	0.001	1.32
Standoff Height*Number of Passes	0.0559	0.0307	1.82	0.077	1.09

Regression Equation in Uncoded Units

```
ln(Line Width) = 4.87 + 0.0346 UA_MFC - 0.0381 Sheath - 0.1780 Print Speed
                 - 0.295 Standoff Height - 0.1346 Number of Passes + 0.001261 UA_MFC*Sheath
                 + 0.00330 Sheath*Standoff Height + 0.01902 Print Speed*Number of Passes
                 + 0.01708 Standoff Height*Number of Passes
```

Figure 24: Regression Model for the Metal Conductive Ink

To overcome the difficulties with multicollinearity, model transformation was done which involved log-transforming the response and standardizing the variables. Residuals for the transformed dataset are shown in Figure 22. With the assumptions being satisfied, stepwise variable selection was performed to remove the insignificant variables with $\alpha = 0.15$. Figure 23 shows the list of variables left in model with their *p*-values, along with the variation explained by the model i.e. 79.20% and 72.47% for adjusted. The regression model to predict the line width of the printed conductive ink using UA of aerosol process is shown in Figure 24 i.e.

$$\begin{aligned}\log(\text{LineWidth}) &= 4.87 + 0.0346(\text{UA_MFC}) - 0.0381(\text{Sheath}) \\ &-0.1780(\text{PrintSpeed}) - 0.295(\text{StandoffHeight}) \\ &-0.1346(\text{NumberofPasses}) + 0.001261(\text{UA_MFC} * \text{Sheath}) \\ &+0.0033(\text{Sheath} * \text{StandoffHeight}) \\ &+0.01902(\text{PrintSpeed} * \text{NumberofPasses}) \\ &+0.01708(\text{StandoffHeight} * \text{NumberofPasses})\end{aligned}$$

4.3.3. Model Validation

This section discusses the model validation of the printed metal conductive ink, which involves computing the line widths using the regression model developed and optically measuring it for a range of parameters. Figure 25 shows the list of 35 tests performed using UA for the conductive ink. For all of the tests, line widths predicted are within 15% of the actual line width measured optically.

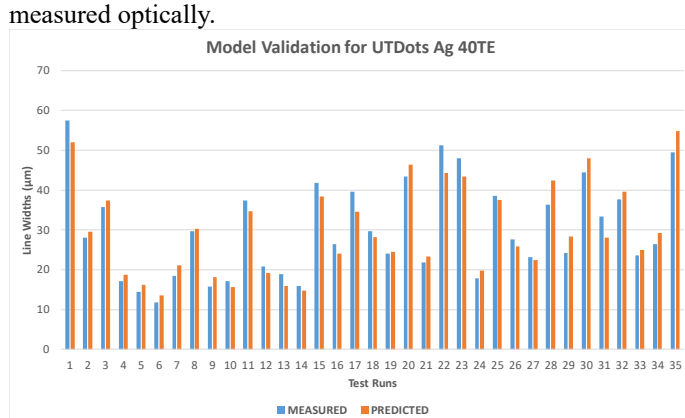

Figure 25: List of 35 tests for Model Validation

4.3.4. Process Capability Statistics

To understand if the aerosol jet printing process is reliable or not, a statistical approach of calculating the process capability

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ratio (C_p) and process capability index (C_{pk}) is implemented. These indices are used to describe the capability of a process by comparing the distribution of the values to the specification limits which the end consumer wants, also knows as a specification target. The equation of these indices are as follows:

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$C_{pk} = \min\left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma}\right)$$

USL:	Upper Specification Limit
LSL:	Lower Specification Limit
μ:	Mean of the distribution
σ:	Standard Deviation

Using these indices, the aim is to find/study if a particular process is achieving the output within the certain specifications. In this paper, the process is aerosol jet printing using pneumatic atomizer with dielectric, and the specification/target was assumed to be $\pm 5\%$ of the desired value. As a general point to note, C_p values are used when the mean of the distribution is aligned with the center width of the specification width (USL–LSL), while C_{pk} is more of a conservative approach, which accounts for the shift, if present. In our case, there is a presence of shift towards left, as can be seen in Figure 26. To account for the shift, C_{pk} value is used, and based on the number of standard deviations which can fit in the distribution, capability can be determined, since 3σ process with $C_p = 1$ is capable of producing 99.75% of products within specification limits.

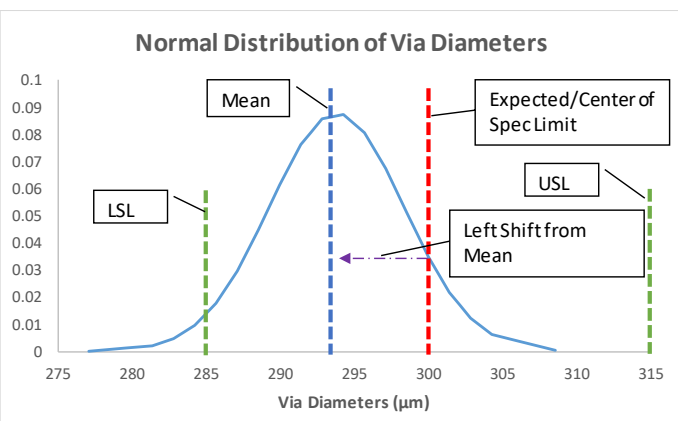

Figure 26: Distribution of Via Diameters

$$C_p = \frac{315 - 285}{6(4.54)} = 1.10$$

$$C_{pk} = \min\left(\frac{315 - 294}{3(4.54)}, \frac{294 - 285}{3(4.54)}\right) = 0.65$$

SUMMARY AND CONCLUSIONS

In this paper, multi-layer print process is developed and different parameters that adversely affect the print quality are studied. Parameters such as UV Curing for the dielectric, sintering temperature for the metal ink, pre-cleaning of the substrate are found to be of great influence to the print quality of the micro-via. Additionally, a statistical approach is implemented for predicting the dimensions such as via diameter by having the process parameters as inputs to obtain the desired characteristics. The approach was used for the two materials individually. After validation, aerosol jet process is gauged as the function of process capability index and ratio to determine if the process is reliable under a specified tolerance.

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